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Database Management Systems

DB101 | Suggested sequence · Semester 2 | Student-authored study guide

How to use these notes

Relational modeling, keys, constraints, normalization, transactions, indexes, and recovery. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Relational model and ER design

Represent entities as relations, rows as tuples, and attributes as columns. A primary key identifies a row; a foreign key enforces a relationship. Model many-to-many relationships with a junction table.

Worked example

```
Enrollment(student_id, course_id, enrolled_on) uses (student_id, course_id) as a composite key
when one enrollment per pair is allowed.
```

Check your understanding

- Why use a junction table?
- What does referential integrity prevent?

MODULE 02 · Normalization and dependencies

Normalization reduces update anomalies by separating facts according to functional dependencies. 1NF uses atomic values; 2NF removes partial dependencies on a composite key; 3NF removes transitive dependencies of non-key attributes.

Worked example

```
If course_id determines course_title, storing course_title on every enrollment duplicates a
fact; move course details to Course.
```

Check your understanding

- Name an insertion anomaly.
- What dependency does 3NF address?

MODULE 03 · Transactions, indexes, and recovery

Transactions group operations. ACID describes atomicity, consistency, isolation, and durability. Indexes speed selected reads at the cost of storage and write work. Use EXPLAIN and workload evidence before adding indexes.

Worked example

```
Transfer: begin; debit one account; credit another; commit only if both updates succeed.
```

Check your understanding

- What is a lost update?
- Why can too many indexes hurt writes?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.